

4.8 Immune Evasion

▼ What is the complement system?

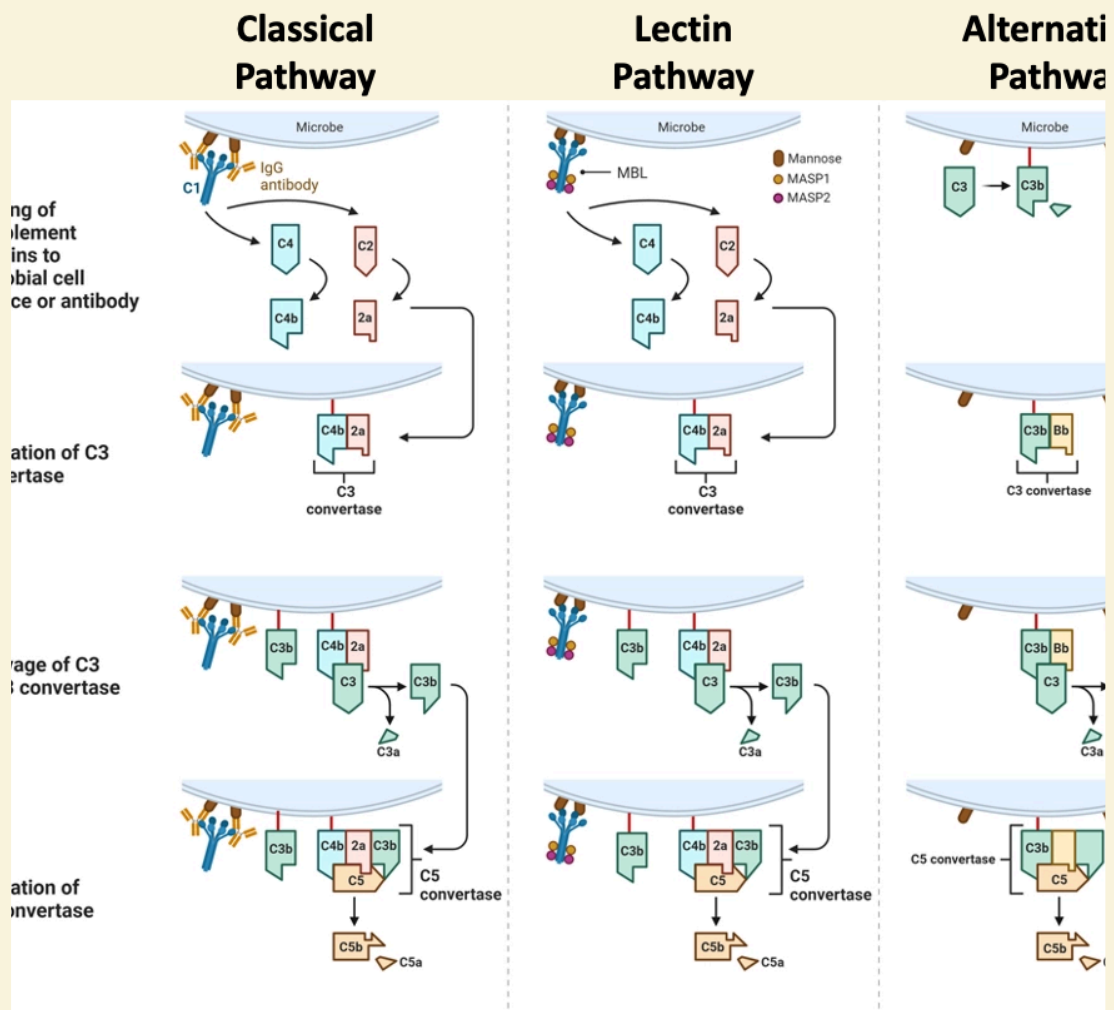
- A group of about 20 plasma proteins crucial to the innate immune response used to opsonise or lyse bacteria.
- Their goal is always to either **opsonise bacteria via C3b** or to **use membrane attack complexes to kill the bacteria**.

▼ What are the three pathways involved in the complement system?

- Classical pathway
- Alternative pathway
- Lectin pathway

▼ Describe the events that occur in the classical pathway.

- Antigen reacts with antibody which activates the **C1 component**
- **C1** breaks down **C4** into **C4a** and **C4b**
 - **C1** combines with **C4b** to form **C14b**
- **C14b** breaks down **C2** into **C2a** and **C2b**
 - **C14b** combines with **C2a** to form **C3 convertase (C14b2a)**
- **C3 convertase** converts **C3** into **C3a** and **C3b**
 - **C3 convertase** combines with **C3b** to form **C5 convertase (C14b2a3b)**
- **C5 convertase** converts **C5** into **C5a** and **C5b**
- **C5b** is used to combine with **C6,C7,C8** and **C9** to form the **Membrane Attack Complex (MAC or C5b6789)**



▼ Which two antibodies can initiate the classical pathway?

- IgM and IgG

- ▼ Describe the events that occur in the lectin pathway.
 - Chemicals called **lectins detect glycans on the surfaces of microbes**
 - This leads to the same cascade (C3 and C5 convertase formation) which ends up in the same response.

- ▼ What is the main difference between the classical and lectin pathways?
 - **Lectin pathway** involves the activation of C4b and C2a via **binding of lectins to sugars**
 - **Classical pathway** involves the activation of C4b and C2a via **binding of antibodies to antigens**

- ▼ What is a bacterial endotoxin?
 - A **lipopolysaccharide** found in the **outer membrane of gram-negative bacteria**, used to **trigger the alternative pathway** of the complement system.

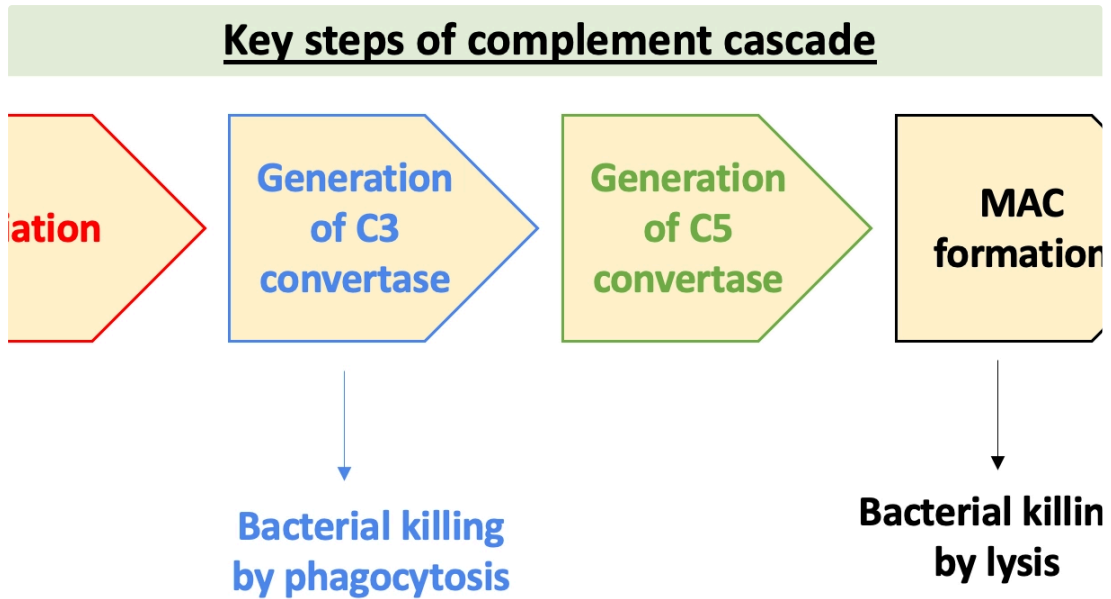
- ▼ Describe the events that occur in the alternative pathway.
 - **C3 is spontaneously hydrolysed to C3b**
 - This occurs continuously, leading to regeneration of C3b.
 - **Factor Bb** binds to **Factor C3b** and this forms a **C3 convertase (C3bBb)**
 - The **C5 convertase is then generated** which allows the MAC to be generated.

- ▼ What role does C3b play in the immune response?
 - It **activates neutrophils and macrophages** which allow them to **phagocytose easier**
 - This is **opsonisation** so C3b can be referred to as an **opsonin**

- ▼ Describe the role of the Membrane Attack Complex (MAC) in the complement system.
 - This **ruptures the bacterial cell wall** which causes **fluid influx** into the bacterial cell and eventual **lysis**.

- ▼ Describe how the MAC is formed from a C5b molecule.
 - **C5 convertase** binds to the bacterial cell wall
 - **C5 convertase** produces **C5b** which also binds to the bacterial cell wall
 - This recruits C6, C7, C8 and C9 to make the MAC

- ▼ Image showing the key steps of the complement cascade.



- Step 2 represents the action of C3b which causes opsonisation of macrophages and neutrophils, leading to phagocytosis

- ▼ What role does C5a play in wider immunity?

- It conducts **chemotaxis** of **neutrophils and macrophages** to the site of infection.

- ▼ What role do factors C3a, C4a and C5a all play in immunity?

- They are involved in the **activation of mast cells and basophils** which leads to the **release of inflammatory mediators** such as **histamine and serotonin**.

- ▼ What is a pyogenic disease?

- A "pus-producing" disease that is characterised by tissue destruction mediated by cytotoxins and hydrolytic enzymes.

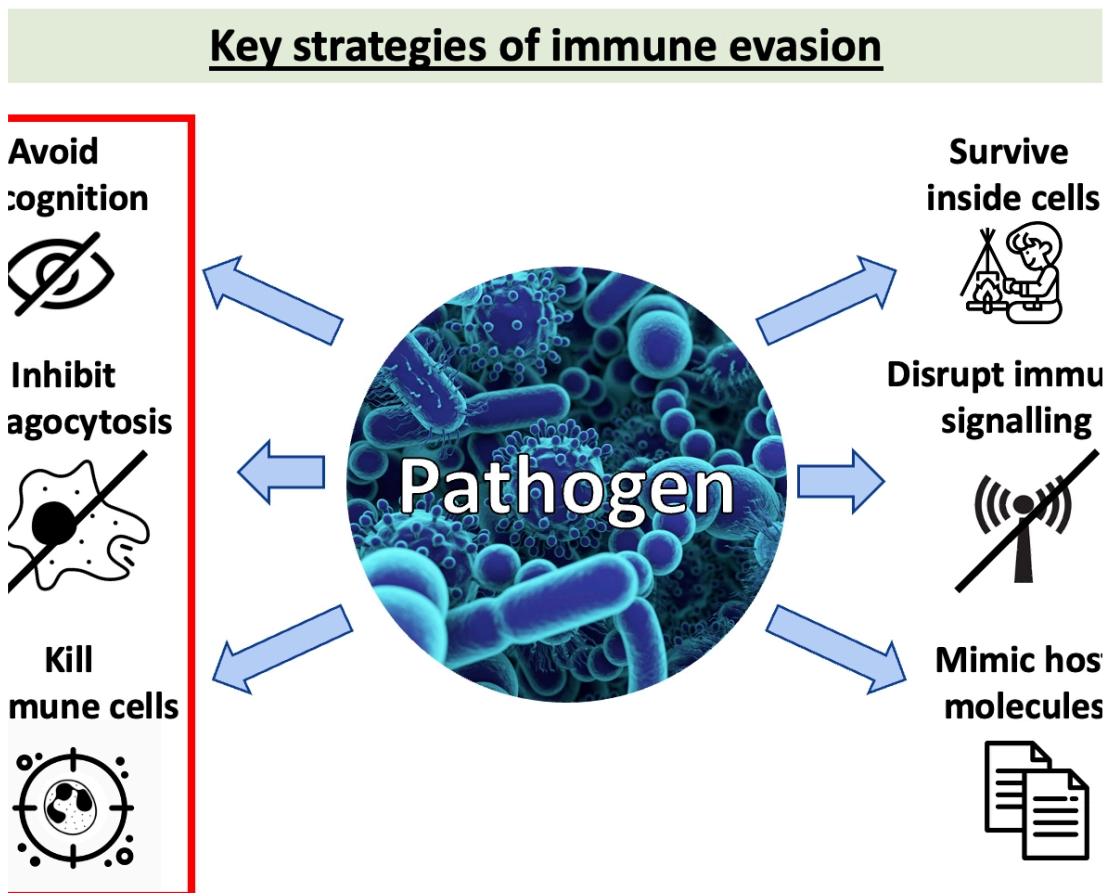
- ▼ Recall the four types of phagocytic cell.

- Neutrophils
- Monocytes
- Macrophages
- Dendritic cells

- ▼ Which two types of molecules can act as opsonins in the immune response?

- C3b
- Antibodies

- ▼ How does C3b interact with phagocytic cells to allow opsonisation of pathogens?
 - C3b binds onto pathogenic surfaces (antigens)
 - The **phagocytic cells can recognise the C3b** using its **complement receptor** and this sends a signal to **begin phagocytosis**
- ▼ How do antibodies interact with phagocytic cells to allow opsonisation of pathogens?
 - Antibodies (IgG/IgM) bind onto the antigen
 - The phagocytic cell uses its **Fc receptor** to recognise the antibody and **initiate a phagocytic response**.
- ▼ Diagram showing the six immune evasion strategies commonly used by pathogens.



- ▼ What type of bacterium is staphylococcus aureus?
 - Gram-positive.
- ▼ What type of infections does staphylococcus aureus cause?
 - Skin and soft tissue infections (i.e bacteraemia and sepsis, endocarditis, osteomyelitis)

▼ What is MRSA?

- Methicillin-resistant staphylococcus aureus.

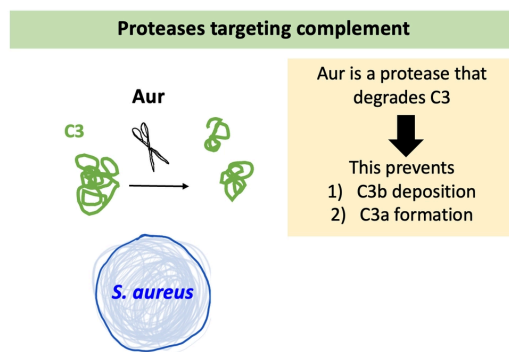
▼ What are the four methods used by bacteria to evade the immune system?

- Degrade C3
- Inhibit C1 binding
- Inhibit C3 and C5 convertases
- Inhibit C5

▼ How does *S. aureus* act directly on complement factor C3?

- It **releases proteases (Aur) that degrade C3** which prevents C3a and C3b formation
- C3b is an important opsonin so the effects of this evasion are increased risk of infection.

***S. aureus* degrades C3**

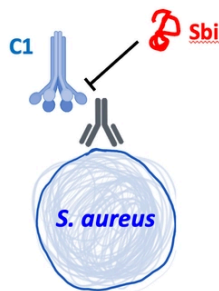


▼ How does *S. aureus* prevent the formation of C3 convertase?

- **Sbi proteins** are released by bacteria which **bind to antibodies**
- This means that **C1 cannot recognise the bacteria** and so there is **no opsonisation**
- C1 not binding to the bacteria means there is a lack of C4 and C2 conversion which is required to form C3 convertase (C4b2a)

S. aureus inhibits formation of C3 convertase

Complement inhibition molecules



Sbi protein binds to IgG to prevent recognition by C1



This prevents formation of the C3 convertase of the classical pathway (CP)

▼ How does *S. aureus* inhibit C3 and C5 convertases?

- It produces SCIN which binds to C3bBb (alternative pathway form of C3 convertase)
- This reduces its enzymatic capacity meaning C3 is not converted which also prevents formation of C5 convertase.

S. aureus inhibits C3 and C5 convertases

Complement inhibition molecules



SCIN protein binds C3bBb



This prevents the formation of C3 convertase and C5 convertase

▼ How does *S. aureus* inhibit C5?

- It releases SSL7 which binds to C5 and **prevent it from converting C5a→C5b** and producing the MAC.

S. aureus inhibits C5

Complement inhibition molecules



SSL7 protein binds C5



This prevents C5 convertases from forming



▼ How does the capsule polysaccharide work in bacterial immune evasion?

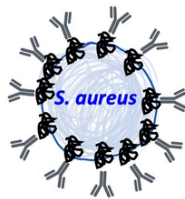
- The thick layer (capsule) forms around the bacteria
- This prevents the antigens from being recognised by the antibodies as they are **hidden**

▼ How do bacteria interact with Fc regions in immune evasion?

- The **bacteria can produce proteins (Protein A) which bind onto the Fc region** of the antibodies that bind to their antigens
 - Normally, antibodies would use their Fab regions to bind pathogens, and Fc receptors in opsonisation to phagocytic cells
- There is obviously less opsonisation as a result of this
- **NOTE:** Antibody-mediated opsonization: In this process, Fab (fragment antigen binding) region of the antibody binds to the antigen, and **the Fc region of the antibody binds to the Fc receptor on the phagocytic cells** and helps phagocytic cells, which include neutrophils, eosinophils, macrophages, and natural killer cells

Surface proteins bind Fc region of antibodies

Expression of antibody binding proteins



Protein A bind antibodies via the Fc region not their Fab region



This prevents normal opsonisation of S. aureus

Other bacteria express surface proteins that bind antibodies including

- *Streptococcus pyogenes* (M protein)
- *Streptococcus dysgalactiae* (Protein G)

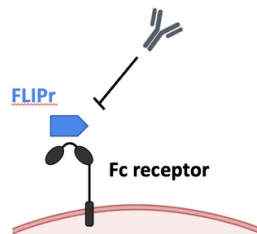
▼ How does staph. aureus cleave antibodies?

- It breaks down **plasminogen into plasmin** which can be used to **digest antibodies**
- Some bacteria directly release the proteases which digest antibodies.

- ▼ How does staph. aureus interact with Fc receptors on phagocytes?
 - It can release molecules such as FLIPr that bind onto the Fc receptor and prevent the opsonisation of bacteria.

Molecules to bind and block Fc receptors

Proteolytic cleavage of antibodies



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- *Streptococci*
- *Neisseria gonorrhoeae* (protease)
- *Haemophilus influenzae* (protease)

- ▼ Diagram summarising how staphylococcus aureus can evade the immune response.

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